Learning to See the Terrain: Semantic Segmentation for Off-Road Navigation

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Improving autonomous navigation through accurate terrain segmentation using deep learning.

Project Summary

Autonomous vehicles require a clear understanding of their surroundings to navigate safely in complex environments. Semantic segmentation enables AI models to classify every pixel in an image into meaningful terrain categories such as trees, rocks, grass, and sky.

In this project, we trained a deep learning segmentation model using synthetic desert environment data generated from Duality AI's Falcon simulation platform. The model uses a DINOv2 Vision Transformer backbone for feature extraction and a ConvNeXt-style segmentation head to classify terrain categories.

Our goal was to train a robust model capable of segmenting unseen desert environments and evaluate its performance using metrics such as IoU (Intersection over Union), Dice score, and pixel accuracy.

Methodology

1. Dataset Description

The dataset used in this project was generated using FalconEditor, Duality AI's digital twin simulation environment.

Dataset Structure

```
dataset/
├── train
│   ├── Color_Images
│   └── Segmentation
├── val
│   ├── Color_Images
│   └── Segmentation
└── testImages
    ├── Color_Images
    └── Segmentation
```

The training set is used to train the model, the validation set is used to monitor performance during training, and the test set is used for final evaluation. Each image has a corresponding segmentation mask where every pixel is labeled with its correct class.

2. Data Preprocessing

Several preprocessing steps were applied before training the model.

- **Image Resizing:** All images were resized to a fixed resolution to ensure consistent input for the model.

- **Normalization:** Images were normalized using ImageNet mean and standard deviation values to stabilize training.

- **Mask Conversion:** Segmentation masks were converted from color-coded labels into numerical class IDs so they could be used for model training.

These preprocessing steps help improve training efficiency and model performance.

3. Model Architecture

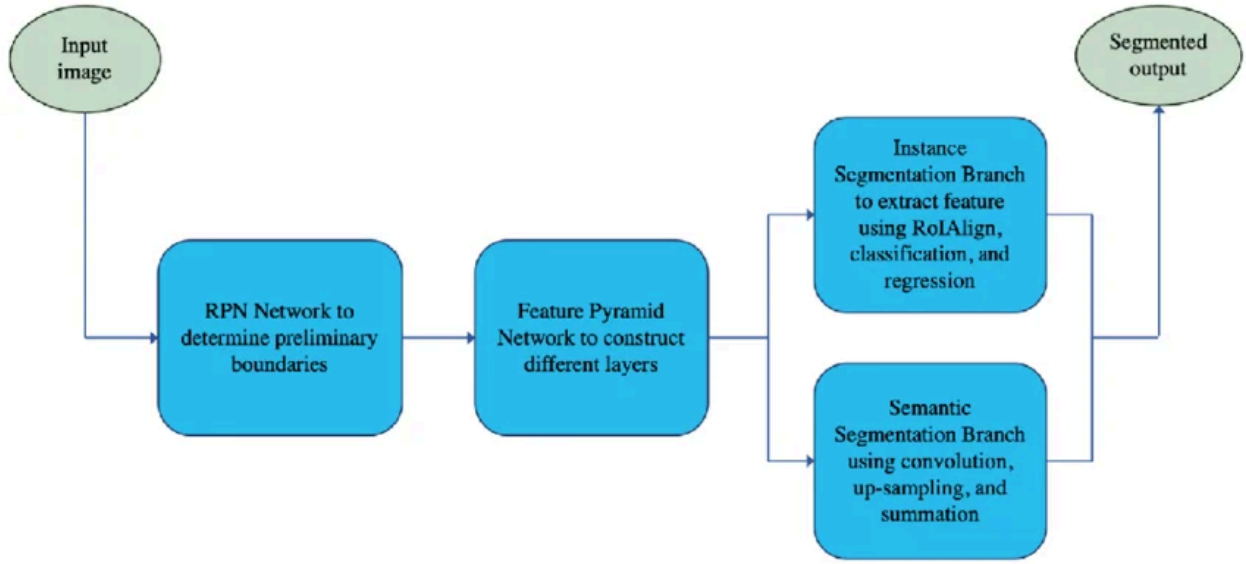
The proposed semantic segmentation model consists of two main components: a feature extraction backbone and a segmentation head.

Feature Extraction Backbone

The model uses the DINOv2 Vision Transformer (ViT) as the backbone. The input image is divided into patches, which are processed using transformer layers to capture global contextual information and extract meaningful visual features.

Segmentation Head

A ConvNeXt-style convolutional segmentation head is used to convert the transformer feature embeddings into pixel-wise predictions. The convolutional layers refine spatial information and generate the final segmentation map, assigning a class label to each pixel in the image.



Model Training

PARAMETER	VALUE
Batch size	**4**
Learning Rate	**0.0001**
Epochs	**25**
Loss function	**Cross Entropy**

Training was performed using PyTorch.

Training Workflow

1. Load training images and segmentation masks
2. Extract patch features using DINOv2
3. Train segmentation head
4. Compute training loss
5. Evaluate model on validation set

Evaluation Metrics

1. IoU (Intersection over Union)

Measures overlap between predicted segmentation and ground truth.

IoU = Intersection / Union

Higher IoU indicates better segmentation accuracy.

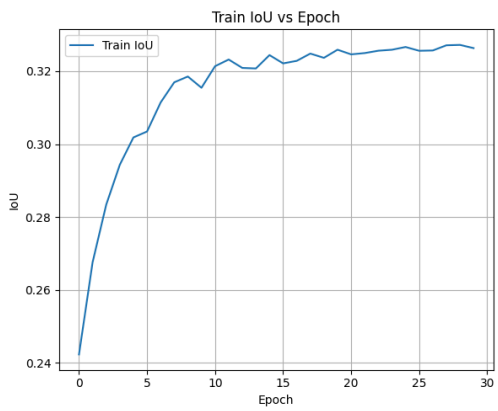
2. Dice Score

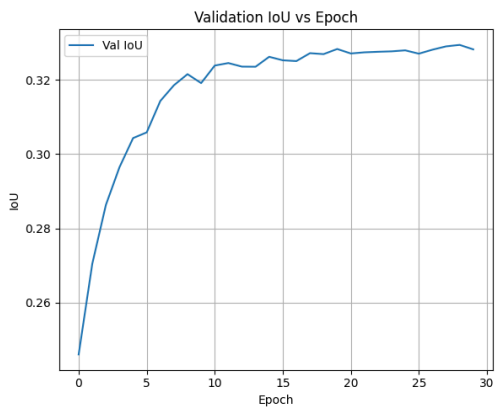
Measures similarity between prediction and ground truth.

Dice = 2 × Intersection / (Prediction + Ground Truth)

3. Pixel Accuracy

Percentage of pixels correctly classified





Challenges and Solutions

1. Similar Appearance of Classes

Several classes in the dataset, such as dry grass, bushes, and ground clutter, have similar colors and textures. This made it difficult for the model to clearly distinguish between them.

Solution:

Data augmentation techniques and a robust semantic segmentation architecture were implemented to help the model learn more discriminative spatial and texture features.

2. Class Imbalance in the Dataset

Certain classes, such as sky and landscape, appeared more frequently than smaller classes like flowers or logs. This imbalance affected the model's ability to accurately detect less frequent objects.

Solution:

A weighted loss function and Dice loss were applied during training to give appropriate importance to minority classes and improve segmentation performance.

3. Inaccurate Boundary Detection

The model initially struggled to correctly classify pixels at object boundaries, particularly between rocks, bushes, and surrounding terrain.

Solution:

Higher-resolution training images and additional training epochs were used to improve the model's ability to learn fine spatial details and produce more precise segmentation boundaries.

4. Overfitting During Training

Early training results indicated that the model performed better on training data compared to validation data, suggesting overfitting.

Solution:

Data augmentation and validation monitoring were incorporated to improve generalization and ensure better performance on unseen images.

5. Computational Constraints

Training deep segmentation models required significant computational resources and longer training time.

Solution:

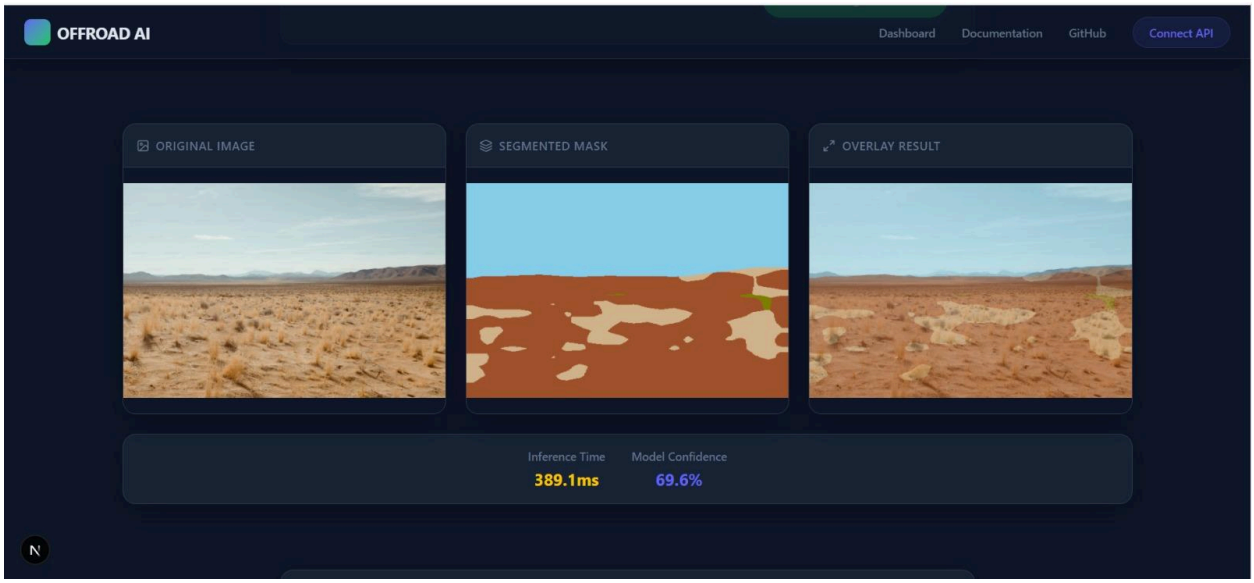
Image resolution and batch size were optimized to maintain a balance between computational efficiency and model performance.

Conclusion

This project focused on developing a semantic segmentation model capable of identifying and classifying different elements present in desert environments. Using the provided dataset, the model was trained to assign a class label to each pixel in an image, enabling accurate scene understanding of objects such as trees, bushes, rocks, dry grass, and sky.

Through data preprocessing, augmentation, and careful model training, the system was able to learn meaningful spatial and visual patterns from the dataset. The performance of the model was evaluated using the Intersection over Union (IoU) metric, which measures the overlap between predicted segmentation and ground truth labels. The results demonstrate that the trained model can effectively segment desert scenes and distinguish between various terrain features.

Overall, the project highlights the effectiveness of deep learning–based semantic segmentation techniques in analyzing complex natural environments and extracting meaningful information from visual data.



Future Work and Potential Improvements

Although the model produced promising results, several improvements could further enhance its performance and applicability.

First, using more advanced segmentation architectures such as transformer-based models or improved convolutional networks could improve feature extraction and increase segmentation accuracy.

Second, increasing the size and diversity of the dataset would help the model generalize better to different environmental conditions, lighting variations, and terrain patterns.

Third, implementing more sophisticated data augmentation strategies could make the model more robust when encountering unseen scenarios.

Additionally, improving boundary detection techniques or incorporating post-processing methods could help reduce misclassification around object edges and further increase the IoU score.

Finally, deploying the model in real-time systems or integrating it with autonomous navigation platforms could extend its practical applications in fields such as environmental monitoring, robotics, and off-road autonomous vehicles.